

Test-Time Structure-Space Search for Antibody–Antigen Structure Prediction

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Abstract

OpenDDE predicts antibody–antigen complexes directly from sequence, but often misplaces the antibody: 34% of retrospective SAbDab targets have a median DockQ below 0.23, while partial interface knowledge from mutagenesis, hydrogen–deuterium exchange, or cross-linking goes unused. We introduce constraint-aware test-time sampling, which uses such knowledge without retraining, a separate docking stage, or gradients through the denoiser. At scheduled denoising steps, a clash-aware refine–search–refine solver rigidly repositions the antibody in the predicted clean structure to satisfy user-specified *Contact* (residue-pair) or *Pocket* (antigen-residue) constraints; sampling then continues from the corrected structure. On held-out SAbDab targets where unguided OpenDDE fails ($N = 805$ for *Contact*, 828 for *Pocket*), top-1 success (DockQ ≥ 0.23) rises from 0.16 to 0.31 with one reference-derived *Contact* constraint and to 0.78 with five, versus 0.27 and 0.33 for constraint-based re-ranking of the same candidates; five *Pocket* residues raise success from 0.17 to 0.33 (re-ranking: 0.26). With the complete epitope, an adaptive policy that guides only when needed improves failed-target success over whole-epitope re-ranking by 0.06–0.18 across six benchmarks, and a guided run takes about 10% longer than an unguided one. These gains require reliable constraints: one wrong *Contact* pair erases them, whereas partial-coverage *Pocket* requests degrade gracefully. Structure-space search at test time thus turns partial but reliable biological knowledge into substantially better co-folding accuracy.

 **GitHub:** <https://github.com/aurekaresearch/OpenDDE>

 **HuggingFace:** <https://huggingface.co/aurekaresearch/OpenDDE>

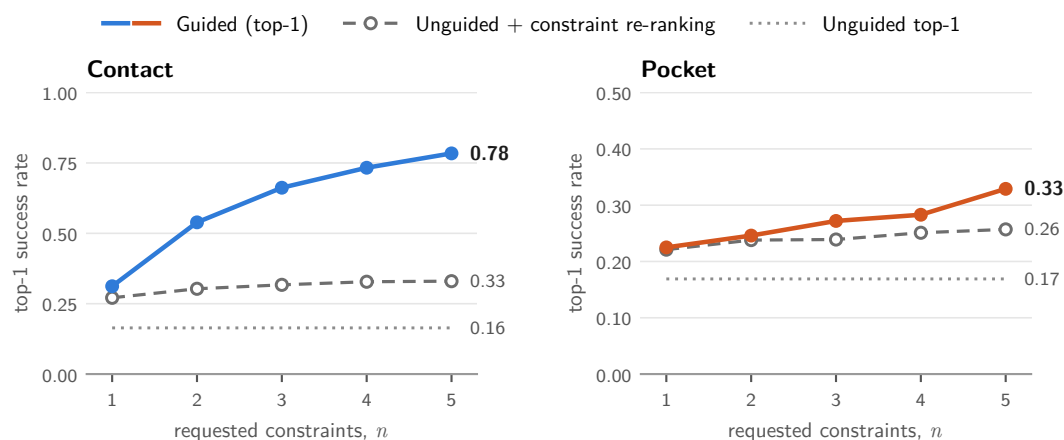


Figure 1: Guided sampling vs. re-ranking: top-1 success on failed SAbDab targets.

*See contributions

Overview

Key Contributions

1. **A training-free test-time sampling operator.** At scheduled denoising steps, the antibody in OpenDDE’s predicted structure is rigidly repositioned by a refine–search–refine solver that reduces constraint violations without creating new severe overlaps (section 1). The network, its weights, and its noise schedule are unchanged.
2. **One solver for two kinds of interface knowledge.** *Contact* (antibody–antigen residue pairs) and *Pocket* (antigen residues of which a required fraction must be touched, antibody side left open) share one energy and one solver. Accelerated evaluation, which matched direct evaluation in all tested pose selections, speeds up pose search by 5,000–56,000 $\times$ and rigid refinement by 46–1,630 $\times$ over direct evaluation, so a guided run takes only about 10% longer than an unguided one (table 1).
3. **Gains beyond re-ranking.** On held-out SAbDab targets where OpenDDE fails, five Contact constraints raise top-1 success from 0.164 to 0.784, versus 0.330 for re-ranking the same unguided candidates; five Pocket residues raise it from 0.169 to 0.329 (section 2).

OpenDDE [1] predicts antibody–antigen complexes directly from sequence, but it frequently misplaces the antibody: on a retrospective SAbDab benchmark [2], 34% of targets have a median DockQ [3] below 0.23 across five samples. Partial interface knowledge is often available from mutagenesis, hydrogen–deuterium exchange, cross-linking, or low-resolution density maps, yet existing ways to use it are costly: restraints encoded as learned model inputs [4, 5] require retraining, a separate docking stage [6] sits outside the co-folding model, and guidance that backpropagates constraint losses through the denoiser [7, 8] is expensive for large models.

We instead spend test-time compute on search in structure space. Antibody placement is a relative-pose problem between two approximately rigid bodies, so we edit the denoiser’s predicted clean structure: we search over rigid-body transformations of the antibody and score each with an inexpensive, clash-aware energy. No retraining, no separate docking pipeline, and no gradients through the model are needed. Partial interface knowledge thus becomes a lightweight way to steer sampling toward plausible binding modes. Running the search inside the sampler is what matters: re-ranking the unguided candidates by constraint satisfaction recovers only about a quarter of the gain from five Contact constraints, and the rest comes from changing the sampling trajectory itself.

1 Method

1.1 Test-Time Sampling with Structure-Space Search

We improve OpenDDE with additional test-time computation and no change to the model. At selected denoising steps, the denoiser predicts a clean antibody–antigen complex. If this prediction satisfies the interface constraints $\mathcal{C}$, it is left unchanged; otherwise we reposition the antibody to reduce the constraint violations and use the repositioned structure in the next sampler update.

Let $x \in \mathbb{R}^{M \times 3}$ be the heavy-atom coordinates of a complex, split into a fixed antigen $\mathcal{A}$ and a movable

antibody $\mathcal{B}$. At denoising step t , the denoiser D_θ maps the noisy state x_t to a predicted clean structure $\hat{x}_0(t) = D_\theta(x_t, t)$, and we move the antibody rigidly:

$$\tilde{x}_0(t) = g_t^* \cdot \hat{x}_0(t), \quad g_t^* = \text{Solve}(\hat{x}_0(t), \mathcal{C}), \quad (1)$$

where $g_t^* = (R, \tau) \in SE(3)$ rotates the antibody about its centroid and translates it. The antigen stays fixed, and both molecules keep their internal geometry. The sampler then takes its usual update step with $\tilde{x}_0(t)$ in place of the denoiser output.

In the main (early) phase we edit the clean estimate $\hat{x}_0$ rather than the noisy state, so that the sampler’s next update starts from a structure with fewer constraint violations and the remaining denoising steps can still reshape the complex around the corrected pose; a short late phase applies the same solver to the sampled state (section 1.4). Each early-phase correction follows one pattern:

Refine $\rightarrow$ Search $\rightarrow$ Refine

A cheap local adjustment comes first, a broad pose search follows only if the constraints still fail, and a final local polish refines the best pose found. Each sample is corrected independently.

1.2 Interface Constraints and Energy

We support two kinds of interface knowledge: *Contact*, which specifies residue pairs, and *Pocket*, which specifies a binding region without residue correspondences. Contact carries precise information; Pocket accommodates partial knowledge of the epitope.

Contact. A Contact constraint pairs an antibody residue in a complementarity-determining region (CDR) with an antigen residue that should interact, and requires their $C\alpha$ distance d_k to lie in $[l, u] = [3.5, 8] \text{ \AA}$. For each pair k , the violation is

$$e_k = \text{ReLU}(d_k - u) + \text{ReLU}(l - d_k), \quad (2)$$

and the constraint set is satisfied when $e_k = 0$ for every pair.

Pocket. A Pocket constraint names n antigen residues that should form the binding interface, without assigning antibody partners. Residue r is *contacted* when its minimum distance to any antibody atom,

$$d_r(x) = \min_{a \in \mathcal{A}_r, b \in \mathcal{B}} \|x_a - x_b\|, \quad (3)$$

is at most $5 \text{ \AA}$, where $\mathcal{A}_r$ contains the atoms of residue r . The request is satisfied when at least K residues are contacted:

$$K = \max(1, \lceil Fn \rceil), \quad d_{(K)} \leq 5 \text{ \AA}, \quad (4)$$

with $F \in (0, 1]$ the required contact fraction and $d_{(K)}$ the K -th smallest residue distance. The optimizer works on the K currently closest residues, with violation $e_r = \text{ReLU}(d_r - 4.5 \text{ \AA})$; aiming at $4.5 \text{ \AA}$ leaves a $0.5 \text{ \AA}$ margin inside the $5 \text{ \AA}$ acceptance test. Because only K of the n residues are enforced, a Pocket request tolerates up to $n - K$ residues that the antibody never touches.

Clash avoidance. Satisfying a constraint must not create physically implausible overlaps, so we add a soft steric penalty

$$C(x) = \sum_{a \in \mathcal{A}, b \in \mathcal{B}} \text{ReLU}(0.85(\rho_a + \rho_b) - d_{ab})^2, \quad (5)$$

where ρ_a and ρ_b are van der Waals radii and d_{ab} is the interatomic distance. An overlap is *severe*

when $d_{ab} < 0.75(\rho_a + \rho_b)$; poses containing one are rejected during search, and refinement may not create new ones.

Energy. Both solver stages minimize a single energy,

$$E(x) = \frac{1}{2}\|e\|^2 + 10 C(x), \quad (6)$$

where e stacks the constraint violations (e_k over all pairs for Contact; e_r over the K closest residues for Pocket). Refinement follows the gradient of E , and search uses its value to rank candidate poses.

1.3 Rigid-Body Solver

At each selected step, the solver searches for a better pose only when the constraints are violated.

Constraint-aware rigid-body update

Input: predicted structure $\hat{x}_0$, constraint set $\mathcal{C}$.

1. **Check:** If $\hat{x}_0$ already satisfies $\mathcal{C}$, return the identity transformation.
2. **Refine:** Apply small rigid-body moves that lower E without creating new severe overlaps or worsening the largest existing one.
3. **Search:** If the constraints remain unsatisfied, score a set of candidate poses and keep the lowest-energy one that has no severe overlap and improves on the current pose.
4. **Refine again:** Locally optimize the resulting pose and return the transformation.

The returned pose has no higher energy than the input, but may still violate the constraints.

Local refinement. Refinement is rigid-body gradient descent on E for 40 iterations: the atomic gradient is reduced to a translation of the antibody and a rotation about its centroid, and each move is backtracked until it lowers E without creating new severe overlaps.

Global pose search. When refinement cannot satisfy the constraints, we score a discrete set of antibody placements. For *Contact*, we first superpose the constrained antibody atoms on their antigen partners by Kabsch fitting [9], then sample rotations and stand-offs around this alignment:

$$25 \text{ approach axes} \times 12 \text{ rotations} \times 5 \text{ stand-offs} = 1,500 \text{ poses.} \quad (7)$$

For *Pocket*, which has no residue pairs, we estimate the antibody's binding face from the atoms currently closest to the antigen and orient it toward the requested epitope:

$$37 \text{ approach axes} \times 24 \text{ spins} \times 5 \text{ stand-offs} = 4,440 \text{ poses.} \quad (8)$$

An epitope spanning several chains uses 24 axes, giving 2,880 poses. Search keeps the lowest-energy candidate without severe overlaps if it improves on the current pose. Local moves fix small placement errors, the global search recovers larger orientation failures, and both preserve the predicted geometry of each molecule.

1.4 Test-Time Sampling Schedule

OpenDDE uses $T = 200$ denoising steps, $t = 0, \dots, 199$, with $t = 0$ corresponding to pure noise. Rather than constraining the whole trajectory, we introduce structure-space search only after meaningful

complex geometry has emerged, in two phases:

$$\begin{aligned}\mathcal{T}_0 &= \{100, 103, \dots, 187\}, \\ \mathcal{T}_1 &= \{190, 192, \dots, 198, 199\}.\end{aligned}\tag{9}$$

In the *early phase* ($t \in \mathcal{T}_0$, 30 steps), the solver corrects the predicted clean structure $\hat{x}_0(t)$ every third step, redirecting the binding pose while diffusion can still reshape the complex around it. In the *late phase* ($t \in \mathcal{T}_1$, 6 steps), the solver acts on the sampled state after each update, mostly refining the pose; the last call also corrects the final output. This schedule concentrates test-time computation on the main uncertainty in antibody–antigen co-folding: the placement of the antibody on the antigen.

1.5 Implementation

Efficient evaluation. Rigid-body search and refinement dominate the cost of guidance. We therefore evaluate them with accelerated algorithms designed to reproduce the decisions of direct evaluation (table 1). The gain has three sources. First, the steric penalty $C(x)$ vanishes beyond a few Å, so a spatial grid restricts its evaluation to nearby atom pairs instead of all antibody–antigen pairs. Second, lower and upper bounds on the energy of each candidate pose, widened by a floating-point error margin, rule out candidates that cannot be the best, so only a few candidates are re-evaluated with the direct energy. Third, all samples and candidate poses are evaluated in parallel on the GPU, including the ten trial step sizes of each refinement iteration. Under the error model, which we verified by testing rather than proved, the bounds never exclude the direct-evaluation winner. Refinement is numerically equivalent rather than bit-identical. In tests, the selected pose agreed with direct evaluation in all 254 Contact search cases, and the accepted refinement moves agreed in all 400 Contact samples. For a complete Contact-guided trajectory, the runtime falls from 2,920 s to 179 s; top-1 success on 90 targets is 0.778 with both implementations (three targets gain and three lose success), and a guided run takes about 10% longer than an unguided one. For Pocket, the accelerated solver made the same decision as direct evaluation in all 11,760 tested sample calls.

Table 1: Runtime of direct and accelerated evaluation on one A100 GPU, per call, on synthetic systems with 900 to 6,000 antibody atoms and 2,800 to 20,000 antigen atoms (five samples; 1,500 candidate poses for Contact and 4,440 for Pocket). The steric row gives the time of one evaluation for 1,500 to 6,000 atoms.

Stage	Mode	Direct	Accelerated	Speed-up
Pose search	Contact	41–791 s	6.8–122 ms	6,100–29,000×
	Pocket	161–5,274 s	32–95 ms	5,000–56,000×
Rigid refinement	Contact	2.3–16.6 s	7.8–23 ms	104–1,630×
	Pocket	11.9–47.5 s	0.26 s	46–183×
Full solver	Contact	44–927 s	0.34–0.50 s	132–1,850×
	Pocket	134–1,347 s	0.32–0.36 s	426–3,759×
Steric potential	Both	1.1–8.0 ms	0.19–0.30 ms	4–34×

Sampler. Guided runs use OpenDDE’s training-free guidance (TFG) sampler, which hosts the rigid-body solver. On protein-only input, its only active term is an inter-chain steric potential, separate from $C(x)$: at every denoising step after the solver call, it moves individual atoms of $\tilde{x}_0(t)$ (not rigid bodies) apart for 20 gradient steps of size 0.01. Unguided runs use the standard sampler without this term. A tiled evaluation accelerates the steric potential 4–34×

Trunk reuse. The constraints enter only the diffusion sampler, not the trunk network. For a given target and random seed, the trunk output is therefore identical across unguided, Contact, and Pocket

runs, so we compute it once and reuse it across sampling conditions. In the primary study, an unguided run takes a median 176 s and the first guided run of a target and seed 188 s, both including the trunk; later guided runs that reuse the trunk take 134 s, or $0.77\times$ a full unguided run. Repeated unguided runs could reuse the trunk in the same way.

2 Experiments

2.1 Experimental Setup

We ask three practical questions. **1.** Does constraint-aware test-time sampling improve antibody–antigen prediction where OpenDDE fails? **2.** How does the gain scale with the amount of interface information? **3.** And how well does it hold up when that information is wrong? Throughout, we separate gains from *generating better structures* from gains available by *re-ranking* the unguided candidates under the same constraints.

Data and split. We start from 3,912 SAbDab complexes released up to September 30, 2021 and keep the 1,328 (34%) on which OpenDDE misplaces the antibody: median DockQ below 0.23 across five unguided samples from seed 101. To prevent leakage, we split these targets into development and test sets by antigen sequence cluster before any tuning. After rebuilding references with collapsed insertion codes and removing references with overlapping chains, 1,305 targets remain. The primary test set contains 866 of them, whose antibody is not identical to, and whose antigen shares little sequence similarity with, any development target. All parameters were selected on development data and frozen before test evaluation.

Interface constraints. Constraints are derived from the reference structures, so they are correct by construction and measure the value of *accurate* interface information, not performance under experimental noise (see [section 2.3](#)). *Contact*: the n closest CDR–antigen $C\alpha$ residue pairs at 3.5–7 Å, each residue used at most once. *Pocket*: the n epitope residues touched by the most antibody heavy atoms within 4.5 Å; a request is satisfied when at least $K = \max(1, \lceil Fn \rceil)$ of them are contacted ([section 1.2](#)), and we use $F = 50\%$, fixed on development data ([section 2.4](#)). We evaluate $n = 1, \dots, 5$. To keep the target set fixed across n , we report all results on targets with at least five available constraints: $N = 805$ targets for Contact and $N = 828$ for Pocket. The unguided baselines therefore differ slightly between the two panels (0.164 vs. 0.169).

Sampling and baselines. All runs use 10 trunk cycles and 200 denoising steps. Each target and condition gets 25 candidates (seeds 101–105 with five samples each), from which one prediction is selected. We compare three policies:

- **Unguided (B_0):** generate candidates without constraints; select the top one by model ranking score.
- **Re-rank only (B_{req}):** use the same unguided candidates, but select the structure satisfying the most requested constraints, breaking ties by ranking score. This isolates the benefit of *selection*.
- **Guided (constraint-aware sampling):** incorporate the constraints during diffusion ([section 1](#)) and select among the resulting candidates with the same constraint-based rule as B_{req} .

As a deliberately strong extra baseline, B_1 re-ranks the unguided candidates by coverage of the *entire* reference epitope, so it sees more interface information than the n residues passed to guidance. A prediction is successful when its interface-mean DockQ is at least 0.23.

Statistics. Uncertainty comes from 1,000 nested bootstrap repetitions over antigen clusters, targets, and seeds. Pre-declared tests are Holm-corrected [10]; others are Benjamini–Hochberg-corrected [11].

2.2 Constraint-Aware Sampling Improves Success Beyond Re-ranking

Constraint-aware sampling lifts top-1 success on targets where unguided OpenDDE fails, and by far more than re-ranking the same candidates can (Figure 1, Table 2). Figure 1 plots top-1 success against the number of constraints n for Contact ($N = 805$) and Pocket ($F = 50\%$, $N = 828$). With five correct Contacts, success rises from 0.164 to 0.784, against 0.330 for re-ranking alone; mean top-1 DockQ rises from 0.137 to 0.428 and the medium-quality rate (DockQ ≥ 0.49) from 0.112 to 0.391. Five Pocket residues raise success from 0.169 to 0.329 (re-ranking: 0.257), a smaller gain.

Table 2: Top-1 success versus the number of constraints n on the primary test set (Contact $N = 805$; Pocket $F = 50\%$, $N = 828$). Bold: guided significantly above re-ranking.

n		1	2	3	4	5
Contact	Unguided	0.164	0.164	0.164	0.164	0.164
	Re-rank only	0.271	0.303	0.317	0.328	0.330
	Guided	0.312	0.539	0.662	0.733	0.784
Pocket	Unguided	0.169	0.169	0.169	0.169	0.169
	Re-rank only	0.221	0.238	0.239	0.251	0.257
	Guided	0.225	0.246	0.272	0.283	0.329

Contact. Gains grow steadily with n . A single contact lifts success from 0.164 to 0.312, but re-ranking already captures about 70% of that gain (0.271). From $n = 2$ on, guidance pulls away: its margin over re-ranking widens from +0.236 at $n = 2$ to +0.454 at $n = 5$, while re-ranking saturates at 0.330. Beyond a single contact, most of the improvement therefore comes from changing the sampling trajectory rather than from selecting better structures from a fixed candidate pool.

Pocket. Pocket guidance improves on unguided sampling at every n , by +0.056 at one residue and +0.160 at five. Its margin over re-ranking is smaller: +0.004, +0.008, +0.033, +0.032, and +0.072 for $n = 1, \dots, 5$. The last three are significant after exploratory correction (BH-adjusted $p = 0.019$, 0.029, and 10^{-6}); the first two are not. With a single residue, 99.6% of guided candidates satisfy the request, so satisfying it says little about the binding pose.

Significance. Table 3 lists the pre-declared comparisons. Contact guidance beats unguided sampling at $n = 5$ (+0.620, P1) and already at $n = 1$ (+0.148, P2), and it outperforms Pocket by +0.452 at $n = 5$ (P4, paired on the 784 targets in both subsets); each additional Contact constraint buys +0.114 success versus +0.024 for Pocket (P3, P7). Against B_1 , which sees the whole epitope, five Pocket residues do not differ significantly (P5: +0.029, $p_{\text{Holm}} = 7.7 \times 10^{-2}$), while a single residue falls short (P6).

Stability. Results are stable across seeds: the per-seed standard deviation of success is 0.012–0.018 for Contact and 0.016–0.026 for Pocket. Pooling candidates across the five seeds helps further: at five Contacts, success is 0.725 per seed (five candidates) and 0.784 pooled (25 candidates). Sensitivity analyses that exclude the 108 targets used earlier to validate the implementation (82 of them in the primary test set), define the difficult-target stratum with the five-seed median, or restrict evaluation to clean references give effects of similar direction and magnitude.

Table 3: Pre-declared comparisons with 95% bootstrap intervals and Holm-corrected p -values.

(a) Contact ($N = 805$; $P4$ on the 784 targets in both subsets).				
Contrast		Δ	95% CI	p_{Holm}
P1	$n=5$ vs unguided	+0.620	[+0.582, +0.675]	2.0×10^{-141}
P2	$n=1$ vs unguided	+0.148	[+0.128, +0.194]	4.3×10^{-28}
P3	Gain per additional constraint	+0.114	[+0.102, +0.126]	2.0×10^{-3}
P4	Contact vs Pocket, $n=5$	+0.452	[+0.394, +0.500]	5.9×10^{-83}
(b) Pocket ($F = 50\%$, $N = 828$).				
Contrast		Δ	95% CI	p_{Holm}
P5	$n=5$ vs B_1	+0.029	[+0.018, +0.090]	7.7×10^{-2}
P6	$n=1$ vs B_1	-0.075	[-0.080, -0.020]	1.1×10^{-7}
P7	Gain per additional constraint	+0.024	[+0.018, +0.031]	2.0×10^{-3}

2.3 Robustness to Incorrect Constraints

Reference-derived constraints are the ideal case; real interface data can be incomplete or wrong. We perturb the requested residues in two ways: *wrong* constraints place the antigen residue at least 12 Å from the true epitope, and *shifted* constraints replace it by a neighbor 4–6 Å away.

Table 4: Top-1 success with perturbed interface constraints ($n = 4$; 274–287 test targets). The unperturbed row is for 287 targets; on the 274 targets of the shifted condition it reads 0.74, 0.30, and 0.27.

Perturbation	Contact	Pocket	
		$F = 50\%$	$F = 100\%$
None	0.71	0.29	0.27
One of four wrong	0.11	0.27	0.03
Two of four wrong	0.12	0.18	0.05
All four shifted (4–6 Å)	0.47	0.20	0.15

Contact guidance is brittle to wrong pairs. Replacing one of four correct contacts cuts success from 0.71 to 0.11, below the unguided top-1 rate on the full test set (0.164): every pair carries equal weight, so a single wrong pair can drag the antibody away from the true interface. Shifted pairs hurt far less (0.47), consistent with the tolerance of the 3.5–8 Å contact window.

Pocket guidance degrades gracefully when only partial coverage is required. At $F = 50\%$ ($K = 2$ of 4), one wrong residue costs 0.02 (0.29 to 0.27; not significant) and two cost 0.11 (to 0.18); since only the K nearest residues are enforced, wrong residues need not be touched. At $F = 100\%$ every residue must be touched, and one wrong residue collapses success from 0.27 to 0.03.

The trade-off is clear: Contact buys precision (0.71 at $n = 4$ when correct) at the price of brittleness, while partial-coverage Pocket buys tolerance at the price of a lower ceiling (0.29).

2.4 Ablation Studies

We ablate the sampling schedule, solver design, and constraint settings on the development split. Parameters were selected with candidate-level DockQ (the mean DockQ over all candidates) on seeds 101–103. A modification was adopted only if its confidence interval excluded zero on the positive side, it improved DockQ by at least +0.02, and it cost no more than 0.02 top-1 success. No tested

modification met this bar, so the original defaults stand.

Table 5: *Schedule and solver ablations on the development split (Contact, $n = 4$). Δ is the change in candidate-level DockQ relative to the default (guided 0.322, unguided 0.089).*

Component	Default	Variant	Δ DockQ
Early start	Step 100	Step 60	−0.001
		Step 80	−0.001
		Step 120	−0.010
Early cadence	Every third step	Every step	+0.001
		Every sixth step	−0.003
Phases	Early and late	Late only	−0.113
		Early only	0.000
Solver order	Refine first	Search first	−0.021

The early phase carries the effect. Removing the early phase costs −0.113 DockQ [95% CI −0.141 to −0.089] for Contact and −0.058 for Pocket; the late phase alone recovers roughly half of the guidance effect. Conversely, removing the late phase when early guidance is present changes nothing (0.000). The main benefit therefore comes from correcting the predicted clean structure during denoising, not from adjusting the final sampled state.

The schedule has slack. Starting at step 60 or 80 and acting every step or every sixth step all stay within 0.003 of the default; only a later start (step 120) costs more, and only −0.010.

Solver order. Searching before refining lowers candidate DockQ by 0.021, supporting the refine-search-refine design: inexpensive local corrections should come before broad pose exploration.

Pocket coverage. We fixed $F = 50\%$ before test evaluation. On development data, requiring 75% or 100% raises candidate-level DockQ at $n = 5$ by +0.0195 and +0.014, both below the +0.02 adoption threshold. On the test set, 50% is best at $n = 4$ and $n = 5$ (0.283 and 0.329), ahead of 75% (0.280 and 0.325) and 100% (0.267 and 0.300). At small n , different fractions can give the same K .

Alternative Pocket strategies. Three follow-ups, evaluated on development data only, also fail the adoption rule: re-ranking by requested-residue distances and model confidence (best +0.026, reduced to +0.015 in cross-validation); expanding requested residues into 6 Å or 8 Å surface neighborhoods or sequence windows (−0.008 to +0.006); and restricting the antibody contact region to CDR residues (−0.003). We therefore keep the original solver, schedule, and 50% Pocket coverage.

2.5 Generalization Across Six Benchmarks

Before the constraint-count study, we tested Pocket guidance on six antibody-antigen benchmarks, passing the *complete* reference epitope (typically 15–20 residues) rather than one to five residues, and requiring $K = \lceil n/2 \rceil$ of its n residues to be contacted. We compare an adaptive routing policy, M_A , with two unguided baselines at a fixed budget of 20 candidates per target:

- **Adaptive (M_A):** generate 10 unguided candidates; if none contacts K epitope residues, generate 10 constraint-aware candidates, otherwise 10 more unguided; select the final prediction by epitope coverage. Guided sampling is thus spent only when the first candidates fail to cover the epitope.
- B_0 : 20 unguided candidates, selected by model ranking score (as in [section 2](#)).
- B_1 : 20 unguided candidates, selected by whole-epitope coverage (likewise).

The six evaluation datasets are OpenBench2026 (192 targets) and ARK-AB (155), two complementary antibody–antigen datasets from 2026, the latter also used in the original OpenDDE evaluation [1]; PXMeterBench (437) and AF3Bench (56); SAbDab entries from October 2021 to December 2025 (447); and the retrospective SAbDab set (3,912). As in the primary study, targets are stratified into *failed* and *solved* by the median DockQ of five unguided seed-101 samples. Unlike the primary study, these seed-101 samples are excluded from the candidate pools, which use seeds 102–105 (20 candidates). This experiment thus differs from the primary one in three ways: it uses the complete epitope, adaptive routing, and 20 rather than 25 candidates. Its success rates should therefore not be compared with those of Table 2.

Table 6: *Top-1 success on six whole-epitope benchmarks with 20 candidates per target (B_0 , B_1 , M_A as in section 2.5). Intervals are 95% cluster-bootstrap intervals; rescued and harmed count targets that cross the success threshold with M_A relative to B_1 .*

(a) Failed targets.								
Dataset	N	B_0	B_1	M_A	$M_A - B_1$	95% CI	Rescued	Harmed
OpenBench2026	79	0.089	0.228	0.405	+0.177	[+0.085, +0.284]	14	0
ARK-AB	49	0.204	0.245	0.429	+0.184	[+0.043, +0.328]	10	1
PXMeterBench	218	0.092	0.197	0.261	+0.064	[+0.023, +0.114]	16	2
AF3Bench	37	0.054	0.162	0.243	+0.081	[0.000, +0.214]	3	0
SAbDab (2021–2025)	167	0.132	0.228	0.341	+0.114	[+0.050, +0.175]	19	0
SAbDab (≤ 2021)	1,328	0.147	0.262	0.358	+0.096	[+0.074, +0.121]	136	8

(b) Solved targets.								
Dataset	N	B_0	B_1	M_A	$M_A - B_1$	95% CI	Rescued	Harmed
OpenBench2026	113	0.973	0.991	0.991	0.000	[0.000, 0.000]	0	0
ARK-AB	106	0.991	0.991	0.991	0.000	[0.000, 0.000]	0	0
PXMeterBench	219	0.968	0.977	0.959	−0.018	[−0.061, 0.000]	0	4
AF3Bench	19	0.947	1.000	1.000	0.000	[0.000, 0.000]	0	0
SAbDab (2021–2025)	280	0.975	0.996	0.993	−0.004	[−0.017, 0.000]	0	1
SAbDab (≤ 2021)	2,584	0.978	0.987	0.987	0.000	[−0.001, +0.002]	2	1

Failed targets. M_A beats re-ranking (B_1) on all six benchmarks, by +0.064 to +0.184. The largest gains are on ARK-AB (+0.184) and OpenBench2026 (+0.177); PXMeterBench (+0.064) and AF3Bench (+0.081) improve less, and the AF3Bench interval reaches zero with 37 failed targets from 17 antigen clusters. On the 1,328 retrospective SAbDab failures, M_A rescues 136 and harms 8.

Solved targets. Guidance rarely disturbs targets the model already solves: the net change is at most 0.004 in magnitude on every benchmark except PXMeterBench, where four of 219 solved targets fall below the threshold. All four belong to one antigen family and were already close to the DockQ cutoff; the extra epitope coverage came at the expense of docking accuracy. Satisfying more interface constraints does not always mean a more accurate complex.

Dataset dependence. The variation in gain across benchmarks is not explained by antibody format, antigen size or type, epitope size, chain count, input features, or training-set similarity. On retrospective SAbDab, targets inside and outside OpenDDE’s training set gain similarly (+0.100 and +0.094), so there is no sign of memorization, although the base model’s training data cover this period. The Contact method with the early-phase schedule has been evaluated on SAbDab only.

Computational cost. For these benchmarks, a guided run cost $1.30\times$ an unguided run. With the accelerated implementation used in the primary study, a guided run takes about 10% longer than an unguided one (median 188 s vs. 176 s over 627 paired runs).

Summary. Guiding the sampler with interface constraints helps far more than re-ranking: five correct Contacts raise top-1 success from 0.164 to 0.784 (re-ranking: 0.330), and adaptive Pocket guidance improves failed targets on all six benchmarks. The gains require reliable constraints: one wrong Contact pair erases them, whereas partial-coverage Pocket requests degrade gracefully. All constraints in this study come from reference structures.

Acknowledgements

We appreciate the community for its patience, understanding, and kind words.

Contributions

Aureka AI research team, GPT 6 Astra and Claude 5.1 Fable.

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